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POINT OF VIEW

Generative AI and collaboration: opportunities for cultivating collective intelligence

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Abstract
The transformative potential of artificial intelligence (AI) is reshaping collaboration within organizations, evoking both excitement and concern. As an individual production technology, AI risks fragmenting workflows, isolating workers, and undermining the human-centered collaboration vital for creativity and innovation. However, as a coordination technology, AI holds enormous promise for enhancing collective intelligence. By considering the fundamental processes underlying intelligence in any system—including reasoning, memory, and attention—we can envision ways AI can overcome traditional barriers to effective collaboration to elevate collective intelligence. This paper explores how AI can help clarify team goals and resolve conflicting motives to enhance collective reasoning; provide personalized knowledge assistance, connect complementary expertise, and mitigate biases to augment collective memory; and facilitate attention to priorities and manage asynchronous and synchronous coordination to optimize the use of collective attention. Despite these opportunities, integrating AI into collaboration presents challenges, including potential impacts on trust, cohesion, and ethical concerns. The paper outlines a research agenda to address these challenges and explore AI's role in fostering inclusive, efficient, and innovative collaboration. By leveraging generative AI responsibly, organizations can amplify human abilities, creating synergistic outcomes that redefine the future of work.

Keywords Collective intelligence · Artificial intelligence · Generative AI · Memory · Attention · Reasoning

Generative AI and collaboration: opportunities for cultivating collective intelligence

The rapid advancements in artificial intelligence (AI), particularly the broader use enabled by generative AI, are reshaping how individuals and groups collaborate and work (Burton et al. 2024). While these technologies evoke both fear and excitement, their actual impact will likely depend on whether organizations adopt them as replacements for or enhancements to human capabilities (Tsvetkova et al. 2024; Woolley & Gupta 2024). As a production technology—particularly an individual production technology that modularizes jobs—AI can threaten individuals' job security and undermine human collaboration, fragmenting workflows, and isolating workers (Crowston & Malone 1988; Henderson & Cooprider 1990). However, as a coordination technology, generative AI has immense potential to amplify collective intelligence, helping humans overcome cognitive limitations, align efforts, and create synergistic outcomes that are uniquely human (Jia et al. 2023; Raisch & Fomina 2024). By understanding how memory, attention, and reasoning functions co-evolve and adapt in any intelligent system, we can identify opportunities to leverage AI to enhance group problem-solving, creativity, and productivity (Woolley & Gupta 2024).

Fear: AI as a production technology that undermines collaboration

The most immediate concern surrounding AI is its role as a production technology that replaces individual human labor and destroys collaboration (Burton et al. 2024). Generative AI (GenAI), capable of producing human-like text, images, and even software code, raises concerns about job displacement and the fragmentation of teamwork (Lee et al. 2023;

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Parent-Rocheleau & Parker 2022). For example, consider a marketing professional using GenAI to produce campaign materials. Tasks that once required brainstorming sessions, collaborative copywriting, and iterative feedback among team members can now be largely automated, with unknown implications for the quality of what is produced. Similarly, in software development, developers may rely on AI to generate code snippets, troubleshoot bugs, and optimize performance without consulting colleagues. The fear is that while these tools may boost efficiency, they reduce opportunities for interpersonal collaboration and erode shared learning, trust, and social cohesion within teams (Schmutz et al. 2024). It is also unknown whether using AI in the place of human collaboration will enhance creativity and innovation or hinder it by steering users toward more common ideas and solutions (Jia et al. 2023).

The impact of AI use on the fragmentation of work may also reinforce feelings of isolation and alienation. Without human-centered collaboration, workers may feel disconnected from broader organizational goals and from each other. For management scholars, this fear highlights a need to examine how overreliance on AI-based production systems impacts individual memory and learning (Sparrow et al. 2011), group dynamics, worker well-being, and long-term organizational resilience (Bankins et al. 2024; Schmutz et al. 2024).

Vision: AI as a coordination technology that enhances collective intelligence

While fears of AI replacing individual humans' jobs and undermining teamwork could be legitimate in some contexts, AI can also serve as a powerful coordination technology to enhance collaboration and collective intelligence (Shin et al. 2023; Woolley & Gupta 2024). AI can act as a complement rather than a substitute for human abilities, enabling individuals and groups to connect skill sets, align priorities, and solve complex problems in ways that were previously impossible (Grace et al. 2018; Gupta et al. 2023).

The challenge of bounded rationality—the cognitive limitations that prevent humans from processing all relevant information when solving problems (Simon 1955)—is arguably why humans must collaborate. Organizational structure enables collective rationality by coordinating the attentional inputs of contributors (Simon 1944). With the increasing dynamism of the business environment, traditional organizing structures are being replaced by more dynamic forms to facilitate interdependence in complex adaptive systems more effectively (Mayo 2022; Puranam 2021; Raveendran et al. 2020). AI offers new opportunities to transcend the limitations of individual cognition and facilitate collaboration even in the absence of a formal organizational structure. By analyzing vast datasets, identifying patterns, and simulating scenarios, AI can help groups coordinate efforts, align goals, and unlock creative synergies that arise from human interaction. Unlike human workers, AI can rapidly process information across disciplines, overcome biases, and surface connections that would otherwise go unnoticed. This capacity creates opportunities for teams to innovate and solve problems more effectively (Shin et al. 2023).

For instance, imagine a cross-functional team working on a new product design. AI can analyze market trends, customer feedback, and competitor data to suggest promising design directions. It can identify potential collaborators across disciplines whose expertise complements the team's strengths. Notably, synthesizing insights and making decisions remain a human responsibility, enabling uniquely human traits like creativity, empathy, and intuition to flourish.

Generative AI and the cognitive foundations of collective intelligence

Models of intelligence across many domains—biological, technological, and hybrid—consistently demonstrate an essential role for three core processes: attention, memory, and reasoning. For instance, in human brains, selective/executive attention routes information and sets control signals (Desimone & Duncan 1995; Petersen & Posner 2012), memory provides both a limited-capacity working space and long-term knowledge (Baddeley 2012; Squire 2004; Tulving 1985), and reasoning transforms these representations via model-based or probabilistic inference (Johnson-Laird 2010; Oaksford & Chater 2007), with frontoparietal multiple-demand systems coordinating these functions for flexible intelligence (Duncan 2010). In technological systems, cognitive architectures implement the same triad explicitly: ACT-R couples perceptual/attentional modules to working and long-term memory with production-rule reasoning (Anderson et al. 2004), while SOAR integrates working memory, long-term semantic/episodic stores, and rule-based problem-solving (Laird 2012). Similarly, in hybrid "joint cognitive systems" involving humans interacting with artifacts and AI, the same three processes are distributed across people and tools—attention as coordination and cueing, memory in external representations, and reasoning in shared problem-solving procedures (Clark & Chalmers 1998; Hollnagel & Woods 2005; Hutchins 1995). We build on these observations to theorize ways for AI to enhance collective intelligence by augmenting memory, attention, and/or reasoning processes (Woolley & Gupta 2024).

Reasoning: aligning goals and resolving competing motives

The foundation of reasoning in any system is the identification of desired end states or goals, as they are the reference point for defining problems (represented by the gap between the current state and desired end state) and generating solutions (by identifying the path between the two; Newell & Simon 1961). While groups can and do exhibit collective reasoning (Colman & Gold 2018; Fitzsimons & Finkel 2018), they frequently struggle due to unclear or misaligned goals or conflicting motives (De Dreu et al. 2008; Raveendran et al. 2020; Raven & Rietsema 1957; Weingart 1992), which human leaders often struggle to identify and address. Generative AI has the potential to improve group reasoning by helping teams detect misaligned goals to prompt clarification, a critical step in collaboration (Locke et al. 1981; Raven & Rietsema 1957). AI can facilitate this process by playing a coaching role and prompting discussion, particularly when team member behavior suggests there may be a conflicting understanding of priorities (Gupta et al. 2024). In more traditional manual labor settings of the last century, leaders and team members could directly observe others' work and draw inferences about goals and motivation. However, in the increasingly distributed and digital environment that most professionals work in today, it is difficult for other humans to observe such patterns clearly. However, AI agents can observe digital traces of work activity and convey information to help leaders detect potential misalignment before work gets too far off-track (Woolley & Gupta 2024).

In addition, when a collective struggles to achieve goal alignment as a result of members' opposing values or preferences, large language models (LLMs) can roleplay various stakeholders to pinpoint key concerns and areas of misalignment (Zhou et al. 2025), building on forecasting methods shown to be effective in understanding and predicting human behavior and preferences (Green & Armstrong 2011). This process of surfacing and testing assumptions can help team members achieve greater clarity and alignment among their goals and anticipate challenges before they proceed (Raven & Rietsema 1957; Weingart 1992). LLMs can also help moderate group conversations to identify common ground (Tessler et al. 2024).

By facilitating the process of clarifying goals and priorities, AI can foster a more profound understanding and alignment among team members and significantly enhance collective reasoning. Ensuring alignment is particularly important in multidisciplinary teams where individuals may implicitly prioritize different outcomes due to their functional perspective and expertise (Cronin & Weingart 2007), leading to subtle misalignments that are hard to pinpoint. AI-based reasoning tools can help surface and bridge these divides, enabling more cohesive and effective collaboration.

Memory: expanding knowledge and connecting expertise

The knowledge that individuals and collectives store and retrieve from memory repositories is a key foundation for collective intelligence (Woolley & Gupta 2024). One of AI's most transformative contributions to collaboration has stemmed from its ability to augment individual memory and expertise by storing and drawing connections to additional knowledge. However, it is important to be judicious in searching for external knowledge, as it can inadvertently undermine the use of the knowledge already possessed within a group (Haas & Hansen 2005; Woolley et al. 2013). Thus, AI systems can enhance collective intelligence by selectively augmenting individual expertise and connecting workers to others with complementary skills and knowledge.

1. *Enhancing individual expertise.* AI tools can act as personalized knowledge assistants, providing workers with relevant information, summaries, and insights on demand, connecting new knowledge to existing knowledge. For example, generative AI can analyze research papers, customer feedback, or historical data to surface key findings that expand or contradict what was known previously and help individuals make informed decisions. This enhancement allows workers to operate with an expanded knowledge base, improving their contributions to group work.
2. *Connecting complementary expertise.* AI can facilitate collaboration by identifying individuals with synergistic knowledge and skills. For instance, AI-driven platforms can analyze employee profiles, past projects, and organizational networks to suggest potential collaborators with diverse but complementary perspectives. By surfacing these connections, AI helps organizations overcome common barriers, such as siloed expertise and geographical dispersion.
3. *Overcoming biases and prejudices.* Human biases often hinder aligning work with expertise. For example, individuals may undervalue contributions from colleagues in underrepresented groups or outside their immediate networks. AI can counteract these biases by suggesting a match of tasks to individuals based on their demonstrated skills and knowledge rather than subjective perceptions. AI can also serve as a "devil's advocate" for a group's plan or conclusion to help expose assumptions or blind spots that may have influenced their decisions.

4. *Facilitating cross-disciplinary and cross-cultural collaboration.* Generative AI translates ideas across disciplinary, linguistic, and cultural boundaries. For instance, AI can translate scientific findings into layperson-friendly language, enabling collaboration between researchers and practitioners and the potential to integrate various tools and concepts. Similarly, AI-based language translation tools allow global teams to communicate seamlessly, reducing misunderstandings and fostering inclusivity.

By enhancing collective memory, AI enables groups to harness broader knowledge and expertise, driving more innovative and effective collaboration.

Attention: facilitating asynchronous and synchronous coordination

Attention also plays a pivotal role in intelligence, drawing guidance from the reasoning system to establish priorities for filtering information to guide accomplishment of the most valuable goals (Newell & Simon 1961; Woolley & Gupta 2024). Memory and attention work interdependently in that attention governs what information enters into memory, while the content of memory shapes attention. However, there can be competition between memory and attention as they prioritize different dimensions in guiding collective coordination. A memory-based logic of organizing prioritizes exploitation of current knowledge and expertise, sometimes reducing efficiency due to bottlenecks as tasks pile up waiting for experts; in contrast, an attention-based logic of organizing prioritizes efficiency by maximizing the use of available attention, possibly at the expense of quality by de-prioritizing use of expertise (Woolley & Gupta 2024). At the individual level, some AI tools can create challenges for managing attention by providing too many stimuli to compete for attention. However, AI tools can also help manage individual and collective attention. At the individual level, tools that structure and filter inputs enable individuals to increase personal productivity by managing attention. In collectives, the establishment of clear and aligned goals as part of the collective reasoning process can also help facilitate collective attention by establishing priorities that guide individual and collective workflows. Additional tools to help facilitate contributors' inputs can also minimize interruptions, allowing collaborators to focus on high-value tasks and engage in more profound thought and creativity.

1. *Asynchronous coordination.* AI tools, such as automated summarizers, schedulers, and project management assistants, can help teams collaborate asynchronously, reducing interruptions and creating more time for deep, creative work. For instance, AI can facilitate online discussions, synthesize meeting notes, flag key decisions, and suggest action items for team members working in different time zones. These artifacts help reduce the distraction and scheduling burdens of overreliance on live meetings and allow workers to engage with tasks on their own schedules.
2. *Synchronous coordination.* When synchronous interaction is necessary—such as in establishing new collaborations or committing to major decisions—AI can optimize coordination. For example, AI tools can analyze team communication to identify gaps or redundancies, ensuring that meetings are focused, efficient, and productive. AI can also facilitate real-time translation and transcription, allowing individuals from different linguistic or cultural backgrounds to collaborate seamlessly.
3. *Alignment of priorities.* Improvement in collective reasoning yields benefits for collective attention due to better alignment of goals and priorities. With shared clarity on objectives, workers can focus on fewer, higher-value projects. In some cases, AI teammates may complete lower-level tasks, such as data entry or content generation, freeing up human bandwidth for more strategic and creative work. This shift allows individuals and groups to engage in deeper, more reflective thinking, fostering innovation.

By alleviating the pressures of multitasking and overscheduling, AI can enable workers to dedicate more attention to meaningful collaboration, improving individual and group performance.

Shaping a research agenda for AI and collaboration

By considering the opportunities to use AI capabilities to enhance coordination by augmenting collective reasoning, memory, and attention, we can envision a future in which AI can significantly enhance collective intelligence. Furthermore, while many of the theorized interventions could be administered via LLMs, it is important to consider ways other forms of AI could be leveraged, particularly in combination with GenAI, to facilitate the processes we describe here. Beyond the NLP capabilities of LLMs, the visualization abilities of GenAI could be helpful for externalizing cognition and representing more complex relationships to facilitate problem-solving; multi-agent architectures could help coordinate across functions and processes, and perhaps facilitate the integration of more traditional AI-based analytics into cross-functional workflows. Thus, as AI capabilities develop,

a better understanding of the mechanisms for enhancing collective cognition and CI will enable organizations to leverage a variety of capabilities either individually or in combination with each other.

Despite the exciting opportunities for AI to enhance CI, many important questions remain that provide fertile ground for future research by organizational scholars across disciplines.

1. *Collective memory.* How can AI systems improve the storage, retrieval, and dissemination of organizational knowledge? What are the implications for knowledge retention and loss when organizations rely on AI for memory?
2. *Attention management.* How does AI help teams prioritize tasks and allocate attention effectively? How does asynchronous and synchronous AI-driven coordination impact team productivity and well-being?
3. *Reasoning and decision-making.* How can AI tools enhance the reasoning processes of groups by facilitating problem definition, trade-off analysis, and perspective-taking? How does AI influence the quality of group decision-making and conflict resolution?
4. *The impact of AI on group dynamics.* How does using AI-based tools influence team trust, communication, and cohesion? Under what conditions does AI enhance or undermine group relationships?
5. *Balancing AI and human contributions.* How can organizations balance automating tasks with AI and preserving opportunities for human-centered collaboration? What tasks are best suited for AI and require human creativity, empathy, and intuition?
6. *AI's role in overcoming biases.* How effective are AI-driven tools in reducing expertise, identity, and collaboration biases? What unintended consequences might arise from reliance on AI to allocate work or evaluate contributions?
7. *The ethical implications of AI in collaboration.* What ethical considerations arise when AI tools mediate team interactions, decisions, and outcomes? How can organizations ensure transparency, fairness, and accountability in AI-enhanced collaboration?
8. *The long-term impact on organizational learning.* How does AI influence organizational learning processes, such as knowledge sharing, innovation, and collective problem-solving? Does reliance on AI reduce or enhance human capabilities over time?

By addressing these questions, scholars can help organizations design systems that leverage generative AI to enhance, rather than replace, human collaboration.

Conclusion

The continuing growth in AI capabilities represents both a challenge and an opportunity for collaboration in organizations. As a production technology, AI risks fragmenting workflows and isolating workers (Bankins et al. 2024; Selenko et al. 2022). However, as a coordination technology, AI has the potential to enhance collective intelligence by improving collective reasoning, attention, and memory (Woolley & Gupta 2024). AI can facilitate more creative, efficient, and inclusive collaboration by helping groups clarify goals, connect expertise, and focus on priorities. For scholars and practitioners alike, the challenge is ensuring that AI complements human abilities, enabling teams to achieve synergies between the two. As AI advances, research on its implications for collaboration will be critical to shaping its role in organizations and society.

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Declarations

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